

SNORKELING

A simple way to observe life underwater is to snorkel. The snorkel goes under the strap of the face mask and sticks out above the water. By breathing in through the mouthpiece, air is drawn down the snorkel and air is expelled through the mouthpiece.



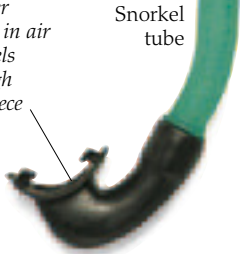
Diver looking at grouper fish in the Red Sea



Face mask traps air to let swimmer view life in the water

Swimmer breathes in air and expels it through mouthpiece

Snorkel tube



Air expelled through end of snorkel



Fins propel swimmer along, but arms should be kept near the body for streamlining



SCUBA DIVING

Use of scuba equipment has proved invaluable in the study of marine life in shallow waters. Instead of bringing animals into an aquarium, marine biologists can observe them in the wild. Sometimes, animals such as hammerhead sharks are sensitive to the noises made by air bubbles and may be scared away.

DEEP STARS

Many different submersibles have been used for underwater exploration (right). The deepest dive ever was to 35,800 ft (10,912 m) in the Mariana Trench by the Swiss engineer Jacques Piccard and US Navy Lieutenant Don Walsh in their bathyscaphe in 1960.



Deep Star can reach depths of 4,000 ft (1,200 m)

Fins used in snorkelling and SCUBA diving

AUTOSUB

This Autonomous Underwater Vehicle (AUV) is capable of exploring remote parts of the ocean. AUVs are unmanned and operate without being tethered to a ship or submersible. Since its launch in July 1996, Autosub has been used in over 300 projects, including missions beneath Antarctic sea ice.

Autosub uses a suite of sensors to collect data

Mechanical arm used to lower Autosub into the water

